

Fundamental constants

Constant	Symbol	Value SI	CGS
Speed of light	c	$2.997925 \times 10^8 \text{ m s}^{-1}$	$\times 10^{10} \text{ cm s}^{-1}$
Electron charge	e	$1.602192 \times 10^{-19} \text{ C}$ 4.80325	$\times 10^{-20} \text{ e.m.u.}$ $\times 10^{-10} \text{ e.s.u.}$
Electron rest mass	m_e	$9.10956 \times 10^{-31} \text{ kg}$	$\times 10^{-28} \text{ gm}$
Proton rest mass	M_p	$1.67261 \times 10^{-27} \text{ kg}$	$\times 10^{-24} \text{ gm}$
Neutron rest mass	M_n	$1.6747 \times 10^{-27} \text{ kg}$	$\times 10^{-24} \text{ gm}$
Planck constant	h $\hbar$	$6.62620 \times 10^{-34} \text{ J s}$ $1.054592 \times 10^{-34} \text{ J s}$	$\times 10^{-27} \text{ erg s}$ $\times 10^{-27} \text{ erg s}$
Charge-to-mass ratio for electron	e/m_e	$1.758803 \times 10^{11} \text{ C kg}^{-1}$	$\times 10^7 \text{ e.m.u. g}^{-1}$
Permeability constant	μ_0	$4\pi \times 10^{-7} \text{ kg m s}^{-2} \text{ A}^{-2}$ 1	— $\times 1 \text{ e.m.u.}$
Molar volume at STP	V_0	$22.4 \times 10^{-3} \text{ m}^3 \text{ mole}^{-1}$	$\times 10^3 \text{ cm}^3 \text{ mole}^{-1}$
Avogadro's number	N_0	$6.022169 \times 10^{26} \text{ kmole}^{-1}$	$\times 10^{23} \text{ mole}^{-1}$
Gas constant	R_0	$8.31434 \times 10^3 \text{ J kmole}^{-1} \text{ K}^{-1}$	$\times 10^7 \text{ erg mole}^{-1} \text{ K}^{-1}$
Boltzmann's constant	k_B	$1.380622 \times 10^{-23} \text{ J K}^{-1}$	$\times 10^{-16} \text{ erg K}^{-1}$
Gravitational constant	G	$6.6732 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$	$\times 10^{-6} \text{ dyne cm}^2 \text{ g}^{-2}$

Bohr magneton $e\hbar/2m_e c$	μ_B	$9.27410 \times 10^{-24} \text{ J T}^{-1}$	$\times 10^{-21} \text{ erg Gauss}^{-1}$
nuclear magneton $e\hbar/2M_p c$	μ_N	$5.05095 \times 10^{-27} \text{ J T}^{-1}$	$\times 10^{-24} \text{ erg Gauss}^{-1}$
gyromagnetic ratio of protons in H_2O	γ_p $\gamma_p/2\pi$	$2.675197 \times 10^8 \text{ rad s}^{-1} \text{ T}^{-1}$ $4.257707 \times 10^7 \text{ Hz T}^{-1}$	$\times 10^4 \text{ rad s}^{-1} \text{ Gauss}^{-1}$ $\times 10^3 \text{ Hz Gauss}^{-1}$

$$\mu_x = g_x \mu_N \sqrt{I(I+1)} = \gamma_x \hbar \sqrt{I(I+1)}$$

$$\mu_x/\mu_N \text{ for } ^1\text{H} = 4.83724$$

The magnetic field for 100 MHz for proton is

$$H = 2.34868 \text{ Tesla}$$

$$1 \text{ Tesla} = 10^4 \text{ Gauss}$$

$$1 \text{ eV} = 1.60219 \times 10^{-12} \text{ erg}$$

$$1 \text{ eV} = 23.05 \text{ kcal mole}^{-1}$$

$$1 \text{ bar} = 14.5038 \text{ psi} = 10^6 \text{ dynes/cm}^2$$

$$1 \text{ atm} = 1.01325 \text{ bar}$$

$$1 \text{ GPa} = 10 \text{ kbar}$$

$$1 \text{ Pa} = 1 \text{ Newton/m}^2 = 10 \text{ dynes/cm}^2$$

$$1 \text{ cP} = 10^{-2} \text{ gm cm}^{-1} \text{ s}^{-1} = 1 \text{ mPa s}$$

$$1 \text{ cSt} = 10^{-2} \text{ cm}^2/\text{s} = 1 \text{ mm}^2/\text{s}$$

$$1 \text{ barrel} = 42 \text{ gal}$$

$$1 \text{ gal} = 3.785412 \times 10^{-3} \text{ m}^3$$

$$1 \text{ Darcy} = 9.869 \times 10^{-9} \text{ cm}^2$$